Pythagorean Theorem

$$\boxed{a^2 + b^2 = c^2}$$

where c is the hypotenuse

This is a Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents.

Now,

The next equation has no integer solution.

$$x^n + y^n = z^n$$

Standard LaTeX practice is to write inline math by enclosing it between `\(...\)`:

> In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Instead if writing (enclosing) inline math between `\(...\)` you can use `$...$` to achieve the same result:

> In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

Or, you can use `\begin{math}...\end{math}`:

> In physics, the mass-energy equivalence is stated by the equation $E = mc^2$, discovered in 1905 by Albert Einstein.

The mass-energy equivalence is described by the famous equation

$$E = mc^2$$

discovered in 1905 by Albert Einstein. In natural units ($c = 1$), the formula expresses the identity

$$E = m \tag{1}$$

This is a simple math expression $\sqrt{x^2 + 1}$ inside text. And this is also the same: $\sqrt{x^2 + 1}$ but by using another command.

This is a simple math expression without numbering

$$\sqrt{x^2 + 1}$$

separated from text.

This is also the same:

$$\sqrt{x^2 + 1}$$

. . . and this:

$$\sqrt{x^2 + 1}$$

The Greek letters are α, β, γ, ρ, σ, δ, ϵ.

$$\alpha + \beta = \gamma$$

$$\rho = \sigma + \delta$$
$$\epsilon = \alpha \times \beta$$